Math 520

Week 6 Homework

Name: Gianluca Crescenzo

Exercise 1. *Let* $H = L^2([0,1])$. *Use the Gram-Schmidt orthogonalization process from linear algebra to produce an orthonormal set* $\{u_1, u_2, u_3\}$ *which spans the same space as* $\operatorname{span}\{1, x, x^2\}$.

Proof. Let $v_1 := 1$, $v_2 := x$, and $v_3 := x^2$. The Gram-Schmidt process gives:

$$r_1 = v_1 = 1,$$
$$\|r_1\| = \sqrt{(r_1, r_1)} = 1.$$

$$r_2 = v_2 - \frac{(v_2, r_1)}{(r_1, r_1)} r_1 = x - \frac{1}{2}.$$
$$\|r_2\| = \sqrt{(r_2, r_2)} = \sqrt{\int_0^1 \left(x - \frac{1}{2}\right)^2 dx} = \frac{1}{\sqrt{12}}$$

$$r_3 = v_3 - \frac{(v_3, r_1)}{(r_1, r_1)} r_1 - \frac{(v_3, r_2)}{(r_2, r_2)} r_2 = x^2 - x + \frac{1}{6},$$
$$\|r_3\| = \sqrt{(r_3, r_3)} = \sqrt{\int_0^1 \left(x^2 - x + \frac{1}{6}\right)^2 dx} = \frac{1}{\sqrt{180}}.$$

Thus an orthonormal basis for $\operatorname{span}\{1, x, x^2\}$ is:

$$u_1 := \frac{r_1}{\|r_1\|} = 1$$
$$u_2 := \frac{r_2}{\|r_2\|} = \sqrt{12}\left(x - \frac{1}{2}\right)$$
$$u_3 := \frac{r_3}{\|r_3\|} = \sqrt{180}\left(x^2 - x + \frac{1}{6}\right).$$

□

Exercise 2. *Show directly that the functions:*

$$u_n(x) = e^{i2\pi n x}, n \in \mathbb{Z}$$

form an orthonormal sequence in $L^1([0,1])$. *Then write the function* $f(x) = x$ *in terms of this basis, that is find* a_n *such that:*

$$x \sim \sum_{n=-\infty}^{\infty} a_n u_n(x).$$

What does Parseval's identity say in this case? (After some algebra you should be able to use your summation to find a formula for $\sum_{n=1}^{\infty} \frac{1}{n^2}$*.)*

Proof. Observe that:

$$\int_0^1 e^{i2\pi n x} e^{-i2\pi n x} dx = \int_0^1 1 dx = 1.$$

$$\begin{aligned}
\int_0^1 e^{i2\pi n x} e^{-i2\pi m x} dx &= \int_0^1 e^{i2\pi(n-m)x} dx \\
&= \left. \frac{e^{i2\pi(n-m)x}}{i2\pi(n-m)} \right|_0^1 \\
&= \frac{e^{i2\pi(n-m)} - 1}{i2\pi(n-m)} \\
&= \frac{\cos\big(2\pi(n-m)\big) - i\sin\big(2\pi(n-m)\big) - 1}{i2\pi(n-m)} \\
&= \frac{1 - 0 - 1}{i2\pi(n-m)} \\
&= 0.
\end{aligned}$$

Thus $\{u_n\}_{n=1}^{\infty}$ is an orthonormal sequence in $L^1([0,1])$. We proceed by cases on computing the Fourier coefficients of x. For $n = 0$:

$$a_0 = \int_0^1 x dx = \frac{1}{2}.$$

For $n \neq 0$:

$$\begin{aligned}
a_n &= (x, u_n) \\
&= \int_0^1 x e^{-i2\pi n x} dx \\
&= -\frac{e^{i2\pi n}(e^{i2\pi n} - i2\pi n - 1)}{4n^2\pi^2} && \text{From Mathematica} \\
&= \frac{i}{2n\pi}. && \text{Also from Mathematica}
\end{aligned}$$

Using Parseval's Identity:

$$
\begin{aligned}
\frac{1}{3} &= \int_0^1 x^2 dx \\
&= \|x\|^2 \\
&= \sum_{n=-\infty}^{\infty} |a_n|^2 \\
&= |a_0|^2 + 2\sum_{n=1}^{\infty} |a_n|^2 \\
&= \frac{1}{4} + 2\sum_{n=1}^{\infty} \left|\frac{i}{2\pi n}\right|^2 \\
&= \frac{1}{4} + \frac{1}{2\pi^2}\sum_{n=1}^{\infty} \frac{1}{n^2}.
\end{aligned}
$$

Solving for $\sum_{n=1}^{\infty} \frac{1}{n^2}$ gives:

$$
\begin{aligned}
\sum_{n=1}^{\infty} \frac{1}{n^2} &= 2\pi^2\left(\frac{1}{3} - \frac{1}{4}\right) \\
&= \frac{2\pi^2}{12} \\
&= \frac{\pi^2}{6}.
\end{aligned}
$$

□

Exercise 3. *Show directly that the functions:*

$$1, \cos x, \sin x, \cos 2x, \sin 2x, \cos 3x, \ldots$$

are mutually orthogonal in $L^2([-\pi, \pi])$.

Proof. For $n, m \neq 0$:

$$
\begin{aligned}
\int_{-\pi}^{\pi} \cos(nx)dx &= \frac{1}{n}\sin(nx)\Big|_{-\pi}^{\pi} = \frac{1}{n}\sin(n\pi) - \frac{1}{n}\sin(-n\pi) = \frac{2}{n}\sin(n\pi) = 0, \\
\int_{-\pi}^{\pi} \sin(nx)dx &= -\frac{1}{n}\cos(nx)\Big|_{-\pi}^{\pi} = -\frac{1}{n}\cos(nx) + \frac{1}{n}\cos(-n\pi) = 0, \\
\int_{-\pi}^{\pi} \cos(nx)\cos(mx)dx &= \int_{-\pi}^{\pi} \frac{\cos((n+m)x) + \cos((n-m)x)}{2} dx = 0, \\
\int_{-\pi}^{\pi} \sin(nx)\sin(mx)dx &= \int_{-\pi}^{\pi} \frac{\cos((n-m)x) - \cos((n+m)x)}{2} dx = 0, \\
\int_{-\pi}^{\pi} \cos(nx)\sin(mx)dx &= \int_{-\pi}^{\pi} \frac{\sin((n+m)x) + \sin((n-m)x)}{2} dx = 0.
\end{aligned}
$$

□

Exercise 4. *Let $\{u_n\}_{n=1}^{\infty}$ be an orthonormal basis for a Hilbert space H. Let $\{v_n\}_{n=1}^{\infty}$ be an orthonormal seq which satisfies $\sum_{n=1}^{\infty} \|u_n - v_n\|^2 < 1$. Show that $\{v_n\}_{n=1}^{\infty}$ is also a Hilbert basis for H.*

Proof. Let $x \in \mathcal{H}$. Suppose $x \perp \{v_j\}_{j=1}^{\infty}$. By Parseval's Identity:

$$\begin{aligned}
\|x\|^2 &= \sum_{j=1}^{\infty} |(x, u_n)|^2 \\
&= \sum_{j=1}^{\infty} |(x, u_n) - (x, v_n)|^2 \\
&= \sum_{j=1}^{\infty} |(x, u_n - v_n)|^2 \\
&\leqslant \sum_{j=1}^{\infty} \|x\|^2 \|u_n - v_n\|^2 .
\end{aligned}$$

Suppose towards contradiction $x \neq 0$. We can divide both sides of the above equation by $\|x\|^2$, giving:

$$1 \leqslant \sum_{j=1}^{\infty} \|u_n - v_n\|^2 < 1.$$

This is a contradiction, hence $x = 0$. Since we've shown $x \perp \{v_j\}_{j=1}^{\infty}$ implies $x = 0$, we've established $\{v_j\}_{j=1}^{\infty}$ as an orthonormal basis for $\mathcal{H}$. □

Exercise 5. *Let H be a Hilbert space and $Y \neq \{0\}$ be a closed subspace, and let P denote orthogonal projection onto Y. Show that:*

(1) P is linear;

(2) P is bounded and $\|P\| = 1$;

(3) For all $x \in H$ we have $(Px, x) = \|Px\|^2$;

(4) $I - P$ is orthogonal projection onto $Y^{\perp}$.

Proof. (a) For any $x_1, x_2 \in H$ and $\alpha \in \mathbb{C}$ we have:

$$\begin{aligned}
P(x_1 + \alpha x_2) &= P(y_1 + y_1^{\perp} + \alpha y_2 + \alpha y_2^{\perp}) \\
&= P(y_1 + \alpha y_2 + (y_1 + \alpha y_2)^{\perp}) \\
&= y_1 + \alpha y_2 \\
&= P(x_1) + \alpha P(x_2).
\end{aligned}$$

Thus T is linear.

(b) Let $x \in H$. Observe that:

$$\begin{aligned}
\|Px\|^2 &= \|y\|^2 \\
&\leqslant \|y\|^2 + \left\|y^\perp\right\|^2 \\
&= \left\|y + y^\perp\right\|^2 \\
&= \|x\|^2 .
\end{aligned}$$

Squaring both sides gives $\|Px\| \leqslant \|x\|$. Diving by $\|x\|$ on both sides of this inequality and taking the supremum over all $x \in H$, $x \neq 0$ gives $\|P\|_{\mathcal{L}(H,Y)} \leqslant 1$. Now let $y \in B_Y(0,1)$. Then:

$$\|Py\| = \|y\| = 1.$$

Hence $1 \in \{\|Px\| \mid x \in B_H(0,1)\}$. It must be the case that $1 \leqslant \|P\|_{\mathcal{L}(H,Y)}$. Therefore $\|P\|_{\mathcal{L}(H,Y)} = 1$.

(c) We will first show that P is self-adjoint and idempotent. Let $x_1, x_2 \in H$. We have:

$$\begin{aligned}
(Px_1, x_2) &= (y, y_2 + y_2^\perp) \\
&= (y_1, y_2) + (y_1, y_2^\perp) \\
&= (y_1, y_2).
\end{aligned}$$

$$\begin{aligned}
(x_1, Px_2) &= (y_1 + y_1^\perp, y_2) \\
&= (y_1, y_2) + (y_1^\perp, y_2) \\
&= (y_1, y_2).
\end{aligned}$$

Therefore $P^* = P$; i.e., P is self-adjoint. Now for any $x \in H$, notice that:

$$\begin{aligned}
P(P(x)) &= P(y) \\
&= y \\
&= P(x).
\end{aligned}$$

Thus $P^2 = P$. Using these two facts:

$$\begin{aligned}
\|Px\|^2 &= (Px, Px) \\
&= (P^*Px, x) \\
&= (P^2x, x) \\
&= (Px, x).
\end{aligned}$$

□

(d) Let $y^\perp \in Y^\perp$. Then $(I - P)(y^\perp) = y^\perp$. Hence $(I - P)^2 = (I - P)$. Now for any $x \in H$:

$$\begin{aligned}
(I - P)(x) &= Ix - Px \\
&= y + y^\perp - P(y + y^\perp) \\
&= y + y^\perp - y \\
&= y^\perp
\end{aligned}$$

Hence $(I - P)$ is an orthogonal projection onto $Y^\perp$.

Exercise 6. *Let* H *be a Hilbert Space with orthonormal basis* $\{u_n\}_{n=1}^{\infty}$*. We try to define a map:*

$$T : H \to H \text{ by } Tx = \sum_{n=1}^{\infty} \lambda_n (x, u_n) u_n$$

for a sequence $(\lambda_n) \subset \mathbb{C}$*. First show that this map is a bounded linear operator if and only if* (λ_n) *is a bounded sequence. Then, under what condition is* T *self-adjoint?*

Proof. Suppose $T \in \mathcal{L}(H)$. Then:

$$\begin{aligned} |\lambda_n| &= |\lambda_n| \, \|u_n\| \\ &= \|\lambda_n u_n\| \\ &= \|Tu_n\| \\ &\leqslant \|T\| \, \|u_n\| \\ &= \|T\| \, . \end{aligned}$$

Taking the supremum over all $n \in \mathbb{N}$ gives $\sup_{n \in \mathbb{N}} |\lambda_n| \leqslant \|T\|_{\mathcal{L}(H)} < \infty$. Therefore (λ_n) is bounded. Conversely, suppose $\sup_{n \in \mathbb{N}} |\lambda_n| := M \leqslant \infty$. We must first show that T is linear. Let $x, y \in H$ and $\alpha \in \mathbb{C}$. Define $a_n := (x, u_n)$ and $b_n := (y, u_n)$. We can write:

$$\begin{aligned} x + \alpha y &= \sum_{n=1}^{\infty} a_n u_n + \alpha \sum_{n=1}^{\infty} b_n u_n \\ &= \sum_{n=1}^{\infty} (a_n + \alpha b_n) u_n. \end{aligned}$$

Applying T to $x + \alpha y$ gives:

$$\begin{aligned} T(x + \alpha y) &= \sum_{n=1}^{\infty} \lambda_n (a_n + \alpha b_n) u_n \\ &= \sum_{n=1}^{\infty} \lambda_n a_n u_n + \alpha \sum_{n=1}^{\infty} \lambda_n b_n u_n \\ &= Tx + \alpha Ty. \end{aligned}$$

Thus T is linear. Now let $x \in X$ and again define $a_n := (x, u_n)$. By Parseval's identity:

$$\begin{aligned}
\|Tx\|^2 &= \sum_{n=1}^{\infty} |(Tx, u_n)|^2 \\
&= \sum_{n=1}^{\infty} \left| \left(\sum_{j=1}^{\infty} \lambda_j a_n u_j, u_n \right) \right|^2 \\
&= \sum_{n=1}^{\infty} \left| \sum_{j=1}^{\infty} \left(\lambda_j a_j u_j, u_n \right) \right|^2 \\
&= \sum_{n=1}^{\infty} |\lambda_n a_n|^2 \\
&\leqslant \sum_{n=1}^{\infty} |M|^2 |a_n|^2 \\
&= |M|^2 \sum_{n=1}^{\infty} |a_n|^2 \\
&= |M|^2 \|x\|^2 .
\end{aligned}$$

Rearranging and taking the square-root of both sides yields:

$$\frac{\|Tx\|}{\|x\|} \leqslant |M|.$$

Taking the supremum over all $x \in X$, $x \neq 0$ gives $\|T\|_{\mathcal{L}(H)} \leqslant |M| < \infty$. Thus $T \in \mathcal{L}(H)$.

Let $x, y \in H$. Observe that:

$$\begin{aligned}
(Tx, y) &= \left(\sum_{n=1}^{\infty} \lambda_n (x, u_n) u_n, y \right) \\
&= \sum_{n=1}^{\infty} \left(\lambda_n (x, u_n) u_n, y \right) \\
&= \sum_{n=1}^{\infty} \lambda_n (x, u_n)(u_n, y).
\end{aligned}$$

$$\begin{aligned}
(x, Ty) &= \left(x, \sum_{n=1}^{\infty} \lambda_n (y, u_n) u_n \right) \\
&= \sum_{n=1}^{\infty} \left(x, \lambda_n (y, u_n) u_n \right) \\
&= \sum_{n=1}^{\infty} \overline{\lambda_n (y, u_n)} (x, u_n) \\
&= \sum_{n=1}^{\infty} \overline{\lambda_n} (x, u_n)(y, u_n).
\end{aligned}$$

From this, T is self-adjoint if and only if $\lambda_n = \overline{\lambda_n}$; i.e., whenever $(\lambda_n) \subset \mathbb{R}$. $\square$